Properties of inequality

1 18 is not greater than 23 or 18 is less than or equal to 23

2

a
i $13 < 16$
ii Yes

b
i $16 > 13$
ii Yes

c
i $-8 < -5$
ii Yes

d
i $7 > 1$
ii Yes

e
i $-4 < -2$
ii Yes

f
i $9 > 7$
ii Yes

g
i $10 < 14$
ii Yes

h
i $-24 < -40$
ii No

i
i $\frac{5}{6} > \frac{2}{6}$
ii Yes

j
i $-\frac{8}{4} < -\frac{12}{4}$
ii No

3 For $a < b$ to be true, a must lie to the left of b on a number line.
Since adding 1 to a number is equivalent to moving that number to the right on a number line, the distance between them on the line will remain the same, so $a + 1$ will still be to the left of $b + 1$.
This same logic applies for $a - 1 < b - 1$, and extends to the general case of $a + n < b + n$ as well.

4
a False **b** False **c** True **d** True

5
a True **b** True **c** False **d** True

6
a Yes **b** No **c** Yes **d** Yes

7
a Yes **b** Yes **c** No **d** Yes

8
a False **b** True **c** True **d** False

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9

$$
\begin{aligned}
a &< b \\
a + a &< b + a \\
a + a + a &< b + a + a \\
3a &< b + a + a \\
3a &< b + b + b \\
3a &< 3b
\end{aligned}
$$

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